



Product Features

- Excellent Insertion Loss and Isolation performance
- High Linearity
- GPIO Control Interface
- Broadband frequency range: 0.1 to 6.0 GHz
- Small package:
- No DC blocking capacitors required
- 1kV HBM ESD Protection on all pins

Product Applications

- 3G/4G/5G multimode cellular tablets and Multi-Mode GSM, EDGE, WCDMA, LTE
- Diversity antenna switching

Product Description

The FM8625H is a Silicon On Insulator (SOI) Single Pole, Double Throw (SPDT) antenna switch which require very low insertion loss, high isolation and high linearity performance.

The high linearity performance and low insertion loss for GSM, EDGE, WCDMA, LTE.

The FM8625H is manufactured in a 1.1 x 0.7 x 0.5mm, 6-pin surface mount Quad Flat No-Lead (DFN) package.

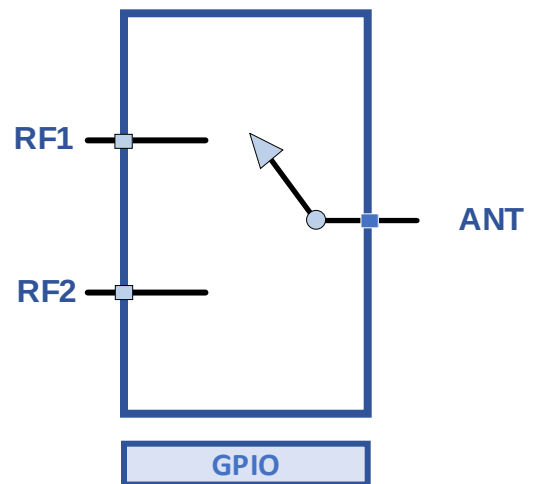


Figure 1 Functional Block Diagram



Absolute Maximum Conditions

Parameters	Symbol	Minimum	Maximum	Units
Supply voltage	V _{DD}	2.5	3.6	V
Control voltage	V _{CC}	0	2.7	V
RF input power	P _{in}		+39	dBm
Storage temperature	T _{STG}	-55	+150	°C
Operating temperature	T _{OP}	-40	+90	°C
Human Body Model, Class 1C	ESD	1000		V

1: Test condition 50% duty cycle, VSWR=1:1, +25 °C

Note: Exposure to maximum rating conditions for extended periods may reduce device reliability. There is no damage to device with only one parameter set at the limit and all other parameters set at or below their nominal value. Exceeding any of the limits listed here may result in permanent damage to the device.

General Electrical Specifications

Parameters	Symbol	Test Condition	Min.	Typ.	Max.	Units
Supply voltage	V _{DD}		2.5	2.8	3.6	V
Supply current, active mode	I _{DD}			50	90	μA
Control signal: High Low			1.35	1.8	2.70 0.3	V
Control current: High Low	I _{CTL}				5	μA
Turn-on time (PIN = +27 dBm)	T _{ON}	Measured from 50% of final VDD supply voltage to 90% of RF power			10	μs
Switching time (PIN = +27 dBm)	T _{SW}	Measured from 50% of final VDD supply voltage to 90% of RF power		2	3	μs

(V_{DD} = 2.85 V, V_{CT} = 1.8 V, T_{OP} = +25 °C, Characteristic Impedance [Z₀] = 50 Ω, Unless Otherwise Noted)



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FM8625H(Document No. S&CIC1682)

SPDT Switch for LTE TRX Application

RF Specifications

Parameters	Symbol	Test Condition	Min.	Typ.	Max.	Units
Operating frequency	f		0.1		6	GHz
Insertion loss	IL	Up to 1.0 GHz Up to 2.0 GHz Up to 3.0 GHz Up to 4.8 GHz Up to 6.0 GHz		0.30 0.35 0.40 0.45 0.65	0.35 0.45 0.50 0.55 0.75	dB
Isolation (ANT port to any receive port)	I _{so}	Up to 1.0 GHz Up to 2.0 GHz Up to 3.0 GHz Up to 4.8 GHz Up to 6.0 GHz	32 27 22 20 12	35 30 25 22 15		dB
2nd Order harmonics	2fo	Pin = +26 dBm,900MHz Pin = +35 dBm,900MHz	-64 -45	-64 -45		dBm
3rd Order harmonics	3fo	Pin = +26 dBm,900MHz Pin = +35 dBm,900MHz	-72 -45	-72 -44		dBm
0.1 dB Compression Point 50% duty cycle, VSWR=1:1	P0.1dB	900M, 50Ω		+39		dBm



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Truth Table

VCTL	ANT-RFX
0	ANT-RF1 on
1	ANT-RF2 on



Pin-out Information

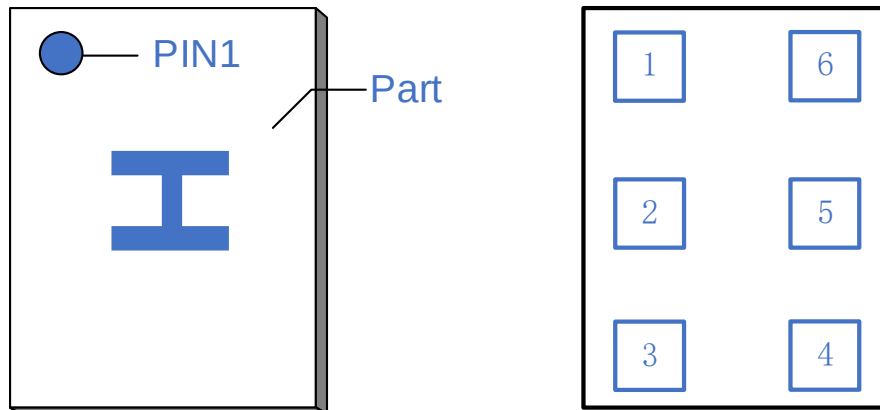


Figure 2 Pin-out Information

Table 1. Pin Description

Pin #	Name	Description	Pin #	Name	Description
1	RF2	RF Port 2	4	VDD	Voltage Supply
2	GND	Ground	5	ANT	Antenna Port
3	RF1	RF Port 1	6	VCTL	Logic Control Voltage



Application circuit

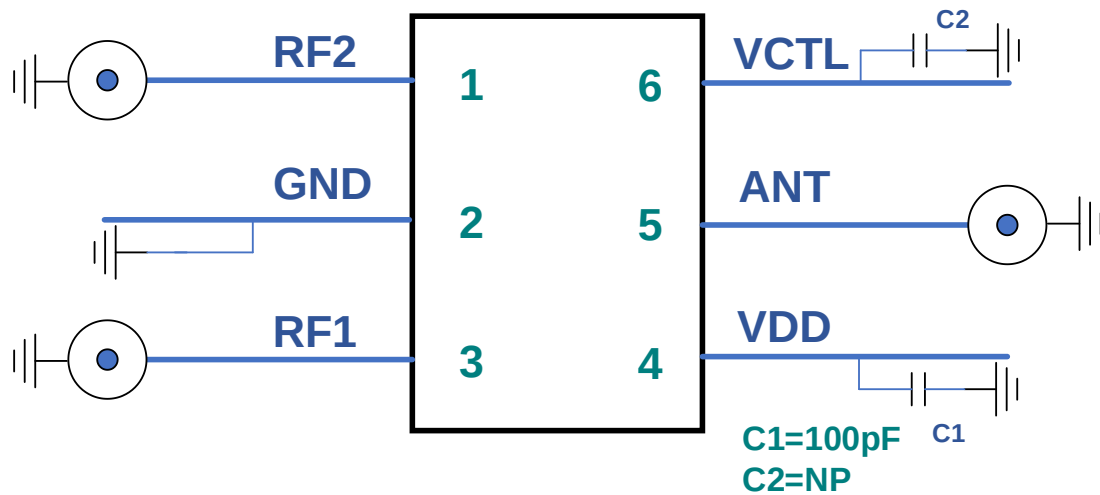


Figure 3 Application circuit



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Evaluation Board

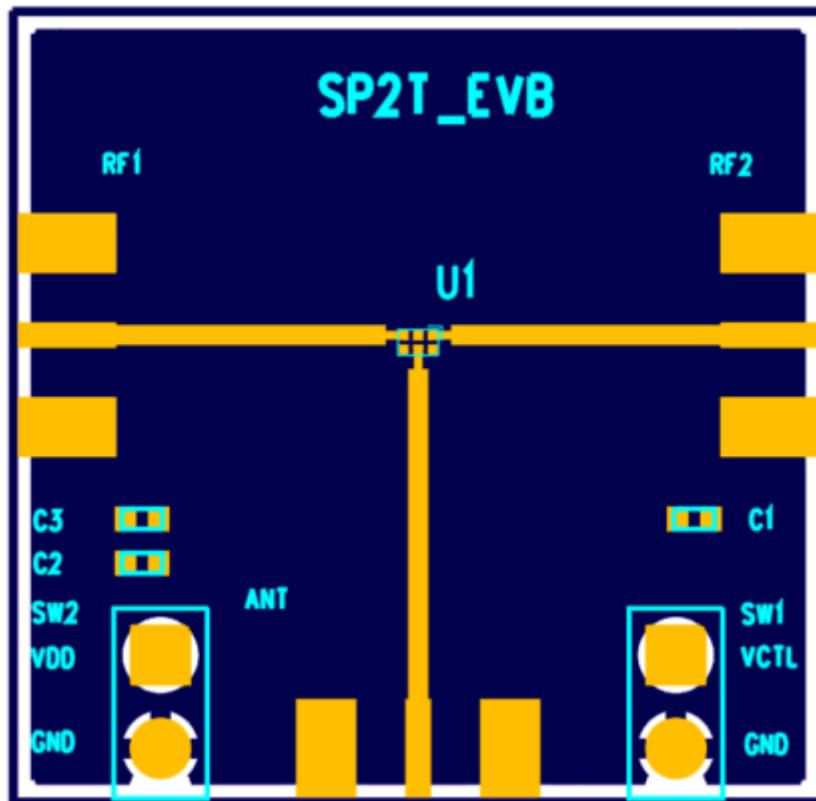
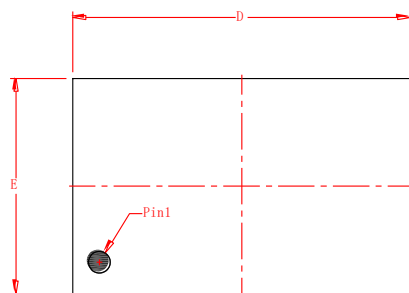


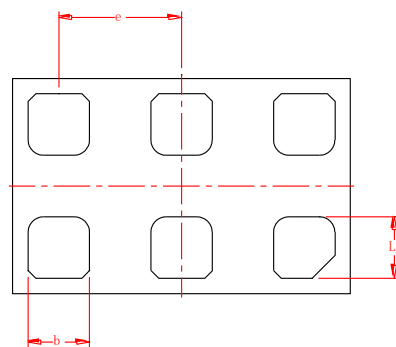
Figure 4 Evaluation Board Assembly Diagram



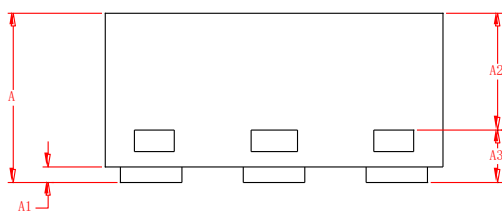
Package Outline Dimension



TOP – VIEW:



BOTTOM – VIEW:



SIDE – VIEW:

SYMBOL	MILLIMETER		
	MIN	NOM	MAX
A	0.45	0.50	0.55
A1	0.00	0.02	0.05
A2	0.33	0.38	0.43
A3	0.127REF.		
b	0.15	0.20	0.25
D	1.05	1.10	1.15
E	0.65	0.70	0.75
e	0.35	0.40	0.45
L	0.15	0.20	0.25

Figure 5 Package Outline Dimension



Package Dimensions (5000pcs)

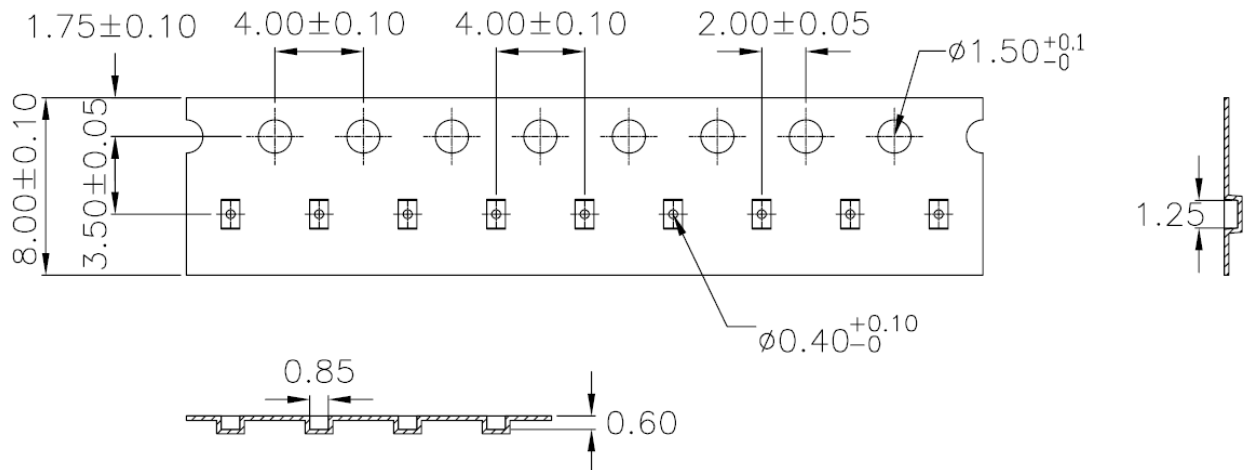


Figure 6 Tape and Reel Dimensions

Declaration of No Harmful Substances

This part is compliant with 2005/20/EC packaging directive, 1907/2006/EC REACH directive and the 2011/65/EU RoHS directive (Restrictions on the Use of Certain Hazardous Substances in Electrical and Electronic Equipment), as amended by Directive 2015/863/EU.

This product also has the following attributes:

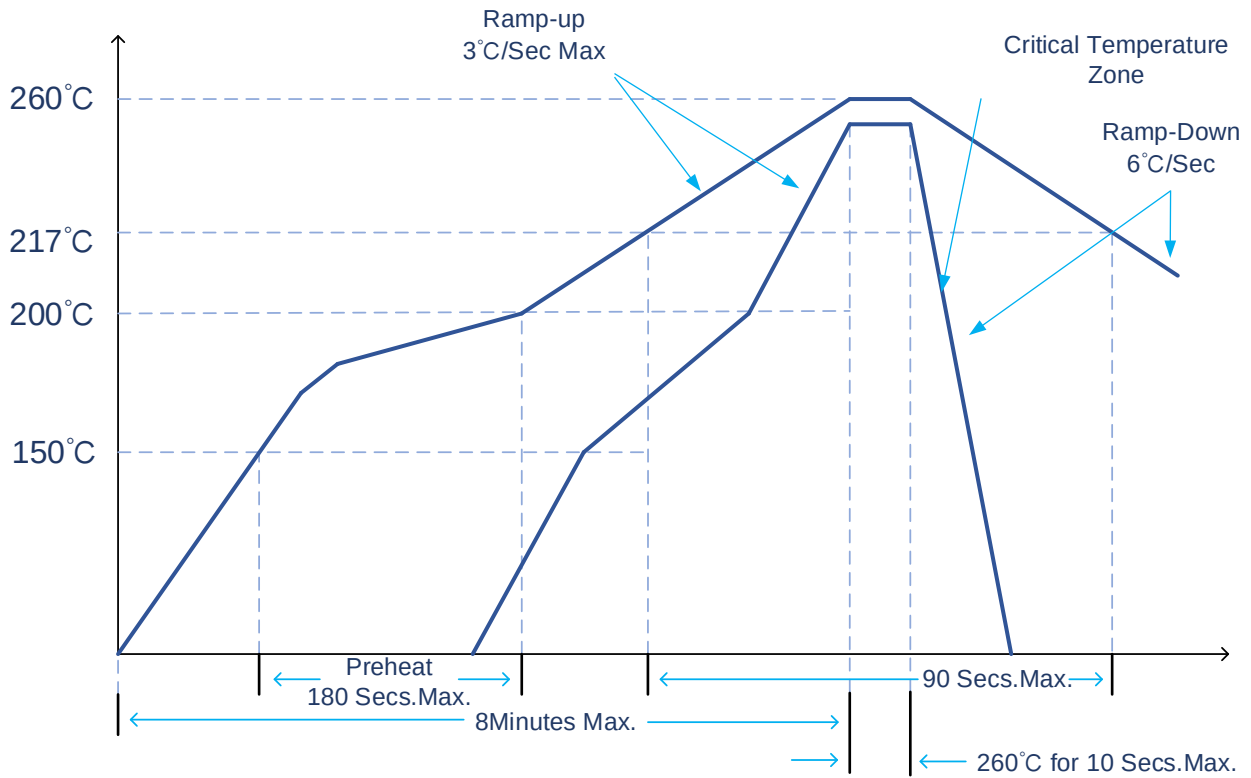
- Lead free
- Halogen Free (Chlorine, Bromine)
- SVHC Free

Version Information

Version	Description	Data	Reviser
V1.3			



Reflow Chart



NOTE: Reflow Profile with 240°C peak also acceptable.